Optimization-based precipitation estimation from commercial microwave links

Thesis defense

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How much is it raining outside?



Photo: NOAA / National Weather Service.

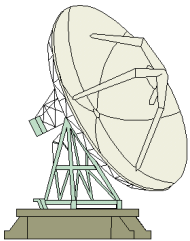
Rain gauges

Collect precipitation at a location and measure its accumulated depth.

Direct point measurements, but sparse coverage between gauges.

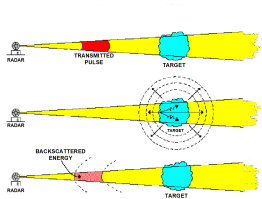
Weather radar

Scanning antenna



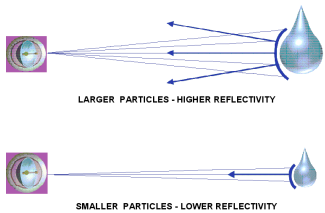
Source: FAA Aviation Weather Handbook, Chapter 15.

Transmitted pulse and echo



Source: FAA Aviation Weather Handbook, Chapter 15.

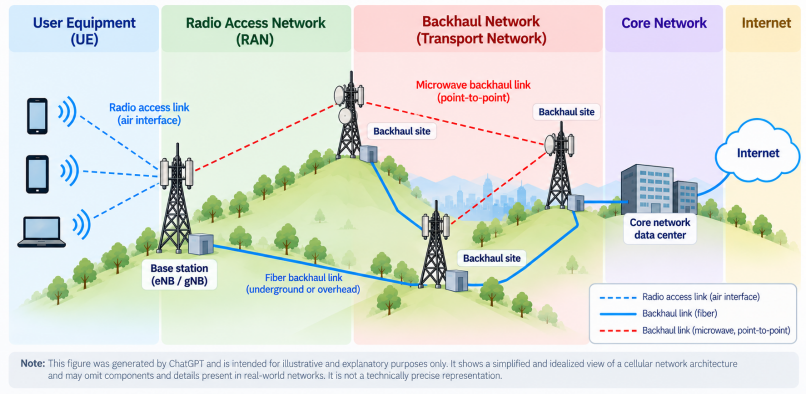
Liquid-target reflectivity



Source: FAA Aviation Weather Handbook, Chapter 15.

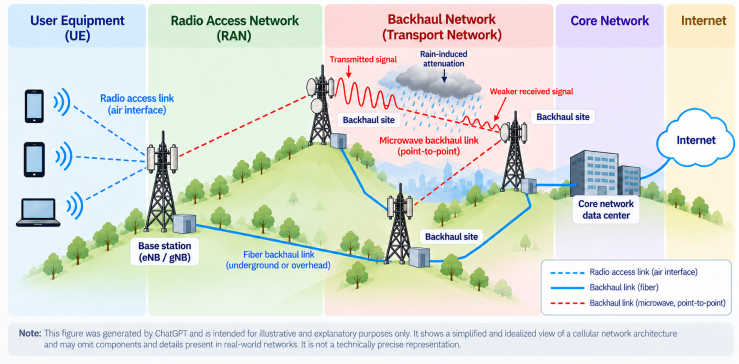
- Reflectivity is converted into an estimated rain rate.
- Broad spatial coverage, but the reflectivity-to-rain-rate conversion introduces uncertainty.

Commercial microwave links in a cellular network



Red dashed connections: point-to-point microwave backhaul.
They carry users' communications.

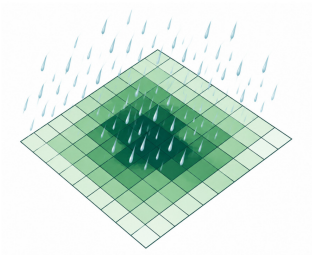
Rain-induced attenuation



- When it rains, the microwave signal loses power.
- This reduction in signal power along the link is called attenuation.
- We use this signal loss to estimate rainfall along the link.

Glossary

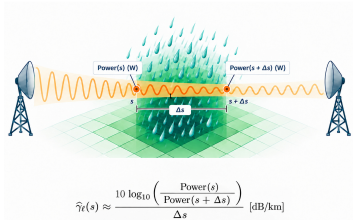
Link ℓ: a line segment joining two geographic endpoints (latitude and longitude), with frequency f_ℓ and polarization p_ℓ.



Conceptual illustration.

Rain field $R(s)$

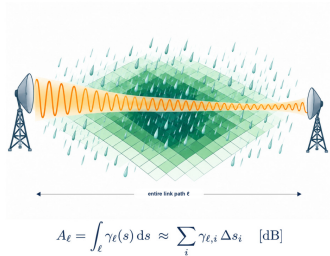
$R : \Omega \to \mathbb{R}_{\geq 0}$. At pixel s, $R(s)$ is rain rate (mm/h).



Conceptual illustration.

Specific attenuation $\gamma_\ell(s)$

Rain-induced signal loss per kilometre along link ℓ (dB/km).



Conceptual illustration.

Total attenuation A_ℓ

Accumulated rain-induced signal loss along the whole link (dB).

Problem Statement

We know: an area Ω containing a set of links $\mathcal{L}$, and their measured rain-induced attenuation A_ℓ for each link $\ell \in \mathcal{L}$.

We want: the unknown rainfall field across the area,

$$R : \Omega \to \mathbb{R}_{\geq 0}, \qquad R(s) \text{ in mm/h}.$$

Goal: use the link paths and their attenuation measurements to estimate the rainfall field $\widehat{R}$.

The rainfall-to-attenuation model

ITU-R P.838-3 maps local rain rate to rain-induced loss per unit distance.

$$\gamma_\ell(s) = \kappa(f_\ell, p_\ell)\, R(s)^{\alpha(f_\ell, p_\ell)}.$$

$R(s)$ in mm/h; $\gamma_\ell(s)$ in dB/km.

Example coefficients (horizontal polarization)

Frequency f_ℓ (GHz)	κ	α
24.92	0.155896	0.999997
30.00	0.240308	0.948457
38.003	0.400172	0.881535

Integrate along a link to obtain total attenuation:

$$\widehat{A}_\ell(R) = \int_\ell \gamma_\ell(s)\, \mathrm{d}s \qquad [\mathrm{dB}].$$

Coefficient model: ITU-R Recommendation P.838-3 (2005).

Milestones in microwave rainfall sensing

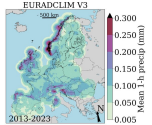
1946–1954	**J. W. Ryde; Gunn & East** Ryde noted near-linearity ($\alpha = 1$); Gunn & East proposed $\gamma_R = kR^\alpha$.
1977–1978	**Atlas & Ulbrich; Olsen et al.** Physical basis: near-linearity at selected frequencies and power-law models.
1991	**Giuli et al.** Tomographic rainfall-field reconstruction from simulated link attenuations.
2000–2003	**Holt et al.; Rahimi et al.** Dedicated dual-frequency links: path rainfall compared with gauges and radar.
2006–2007	**Messer et al.; Leijnse et al.** Rainfall retrieval from commercial links in Israel and the Netherlands.
2013 / 2016	**Overeem et al.** Netherlands-wide maps using ordinary kriging; later, 2.5 years of maps.
2020	**Graf et al.** German-wide evaluation with approximately 4,000 CMLs and IDW.

Our comparison uses IDW, an established baseline in large-scale CML rainfall mapping.

Benchmark construction

From European precipitation maps to synthetic link observations

1 Precipitation fields



84 hourly maps
18–31 Jan 2023
2 km grid

KNMI / EURADCLIM

2 Patch selection



Detect rain events;
select **100 patches**

Basemap: OSM / CARTO

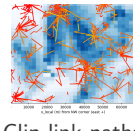
3 Network template



4TU Netherlands
Mean: **1,324**
per patch
Range: 939–1,996

4TU data + OSM / CARTO

4 Synthetic link data

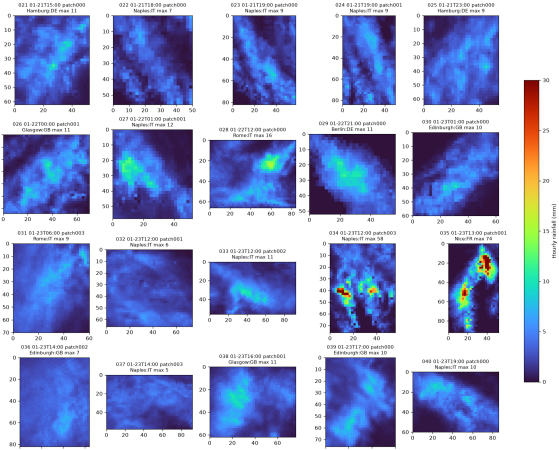


Clip link paths
to the patch
Simulate rain-
induced attenuation

EURADCLIM + 4TU links

We use the 4TU link geometry and frequencies, but not the received signal levels (RSLs). Attenuation is simulated from each reference rainfall field.

Example benchmark patches



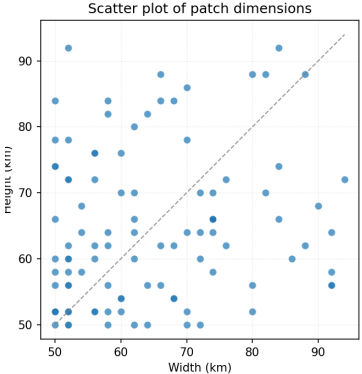
Examples of benchmark rainfall patches from EURADCLIM.

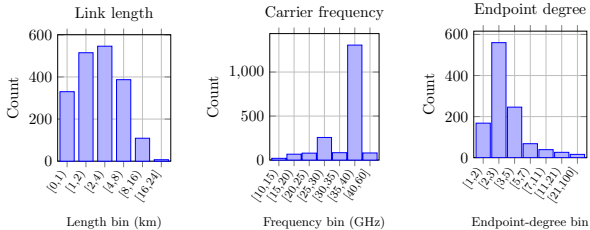
Each panel shows one selected hourly rainfall patch. The color scale is hourly rainfall accumulation (mm); the axes show native grid-cell indices.

The 100-patch benchmark samples 84 hourly EURADCLIM maps from 18–31 January 2023. The source maps have a 2 km grid.

These examples show the range of patch shapes and rainfall patterns used to simulate link attenuations.

Patch geometry and link distributions





Patch geometry: all 100 patches **Link distributions:** largest patch (88×88 km; 1,894 link observations)

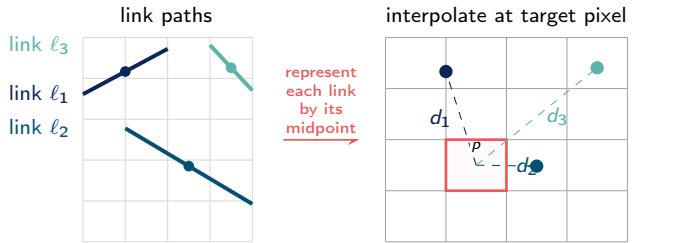
Reconstruction methods

Method	Main idea
IDW	Use link midpoints as virtual rain gauges; weight observations by distance to the target pixel.
ILDW	Weight observations by distance from the target pixel to the link segment.
Optimization	Estimate pixel rain rates by fitting attenuations while controlling spatial variation.

Reconstruction methods

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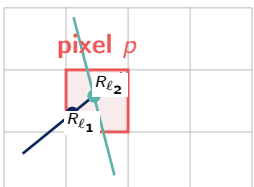
IDW: turn each link into a point observation



$$\widehat{R}(p) = \sum_{\ell \in N_p} w_{p\ell} R_\ell, \qquad w_{p\ell} = \frac{d_{p\ell}^{-2}}{\sum_{j \in N_p} d_{pj}^{-2}}, \qquad d_{p\ell} = \|c_p - m_\ell\|_2.$$

N_p: link midpoints within 15 km of pixel center c_p. Closer midpoint $\Rightarrow$ larger weight.

Discretized IDW: when a pixel contains midpoints



Two link midpoints fall inside the same pixel.

Use the local midpoint observations directly

$$M_p = \{\ell : m_\ell \in p\},$$
$$\widehat{R}_p = \frac{1}{|M_p|} \sum_{\ell \in M_p} R_\ell, \qquad M_p \neq \varnothing.$$

Here R_ℓ is link ℓ's rain-rate estimate. If several midpoints fall in pixel p, average those estimates.

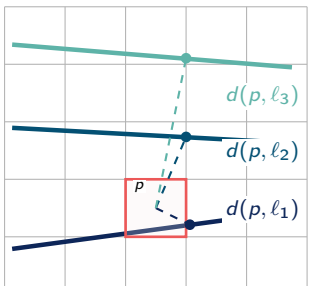
An occupied pixel gets a direct local estimate. Use distance-weighted IDW only when no midpoint falls inside.

Reconstruction methods

Method	Main idea
IDW	Use link midpoints as virtual rain gauges; weight observations by distance to the target pixel.
ILDW	Weight observations by distance from the target pixel to the link segment.
Optimization	Estimate pixel rain rates by fitting attenuations while controlling spatial variation.

(Discretized) ILDW: distance to each link

Measure the shortest distance from pixel center to each link segment.



Same link-level rain estimates and inverse-square rule as IDW; only the distance changes:

$$d_{p\ell} = \mathrm{dist}(c_p, \ell), \qquad w_{p\ell} = \frac{d_{p\ell}^{-2}}{\sum_{j \in N_p} d_{pj}^{-2}}.$$

$$\widehat{R}(p) = \sum_{\ell \in N_p} w_{p\ell} R_\ell.$$

If a link crosses the pixel, average the crossing links; otherwise use links within 15 km.

Reconstruction methods

Method	Main idea
IDW	Treat each link midpoint as a virtual rain gauge.
ILDW	Weight observations by distance to the link segment.
Optimization	Estimate pixel rain rates by fitting attenuations while controlling spatial variation.

Optimization

Using L-BFGS-B, we estimate a nonnegative rain rate in every pixel by minimizing

$$J(r) = \alpha_{\text{atten}} J_{\text{atten}}(r) + \alpha_{1d} J_{1d}(r) + \alpha_{2d} J_{2d}(r) + \alpha_{\text{total}} J_{\text{total}}(r), \qquad r \geq 0.$$

J_{atten}	Fits predicted link attenuations to the observations.
J_{1d}	Penalizes differences between neighbouring pixels.
J_{2d}	Penalizes changes in those differences, discouraging curvature and oscillation.
J_{total}	Penalizes mean squared rain rate, weakly shrinking the field toward zero.

For each patch, the weights use the cost values at the ILDW estimate:

$$\alpha_{\text{atten}} = \frac{1}{J_{\text{atten}}(r_{\text{ILDW}})}, \qquad \alpha_{1d} = \frac{1}{J_{1d}(r_{\text{ILDW}})},$$
$$\alpha_{2d} = \frac{1}{2J_{2d}(r_{\text{ILDW}})}, \qquad \alpha_{\text{total}} = \frac{1}{|P| J_{\text{total}}(r_{\text{ILDW}})}.$$

In our experiments, the J_{total} term did not noticeably affect reconstruction quality.

Problem: nonconvex attenuation fit

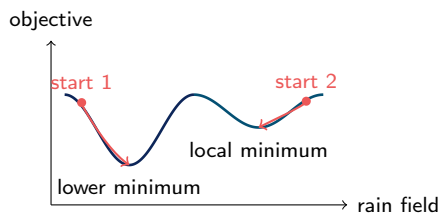
The attenuation-mismatch objective is

$$J_{\mathrm{atten}}(r) = \sum_{\ell} \left(\widehat{A}_\ell(r) - A_\ell\right)^2.$$

With the power-law forward model, this objective **can be nonconvex** when a link's exponent $\alpha_\ell \neq 1$.

Why it matters

- Optimization may settle at a local solution.
- The result may depend on the starting rain field.



Convex surrogate using virtual links

1. Choose a nearly linear reference model. At $f_0 = 24.92$ GHz with horizontal polarization, ITU-R gives $\alpha_0 \approx 1$. In the surrogate, set $\alpha_0 = 1$:

$$\widehat{A}^0_\ell(r) = \kappa_0 \int_\ell r(s)\,\mathrm{d}s = \kappa_0 \sum_p d_{\ell,p} r_p.$$

2. Convert each observation to an equivalent uniform rain rate.

$$r^{\mathrm{eq}}_\ell = \left(\frac{A^{\mathrm{obs}}_\ell / |\ell|}{\kappa_\ell} \right)^{1/\alpha_\ell}, \qquad A^0_\ell = \kappa_0 r^{\mathrm{eq}}_\ell |\ell|.$$

The virtual link keeps the original path geometry and uses the same equivalent rain rate.

3. Fit the virtual-link attenuations.

$$J_0(r) = \sum_\ell \left(\widehat{A}^0_\ell(r) - A^0_\ell \right)^2.$$

Since $\widehat{A}^0_\ell(r)$ is linear in r, this least-squares term is convex. Adding convex smoothness and shrinkage terms preserves convexity; a positive ℓ_2 shrinkage term makes the surrogate strictly convex.

Homotopy continuation

Rather than switch directly to the nonlinear original model, gradually interpolate each link's frequency:

$$f_\ell^\lambda = (1-\lambda) f^0 + \lambda f_\ell, \qquad \lambda = 0, 0.1, \ldots, 1.$$

At each step, warm-start from the previous rain-field estimate



$\lambda = 0$
Convex surrogate
$f_\ell^0 = f^0$

Intermediate models
$\lambda = 0.1, \ldots, 0.9$

$\lambda = 1$
Original link frequencies
$f_\ell^1 = f_\ell$

Continuation reaches the original model, but does not guarantee its global optimum.

Evaluation metrics

Let R_i and $\hat{R}_i$ be the reference and reconstructed rain rates at pixel i, for n paired pixels.

$$\text{Bias} = \frac{1}{n}\sum_{i=1}^{n}(\hat{R}_i - R_i) \qquad \text{mean signed error}$$

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\hat{R}_i - R_i)^2} \qquad \text{root mean squared error}$$

Both metrics have units of mm/h. Positive bias means overestimation.

Pearson correlation. Do above- and below-average rainfall occur in the same locations in both maps?

$$\rho = \frac{\sum_i (\hat{R}_i - \bar{\hat{R}})(R_i - \bar{R})}{\sqrt{\sum_i (\hat{R}_i - \bar{\hat{R}})^2}\sqrt{\sum_i (R_i - \bar{R})^2}}.$$

Whole-patch reconstruction results

Method	RMSE (mm/h)	Signed bias (mm/h)	Pearson r
IDW	1.307 ± 1.234	0.034 ± 0.296	0.866 ± 0.059
ILDW	1.239 ± 1.182	0.047 ± 0.261	0.878 ± 0.055
Solver(ILDW)	1.059 ± 1.192	0.040 ± 0.235	0.918 ± 0.055
Convex Solver	0.845 ± 0.803	0.023 ± 0.075	0.940 ± 0.034
Homotopy Solver	0.845 ± 0.803	0.023 ± 0.075	0.940 ± 0.034
Solver(GT)	0.503 ± 0.369	0.009 ± 0.030	0.961 ± 0.032

Mean $\pm$ standard deviation across 100 patches.

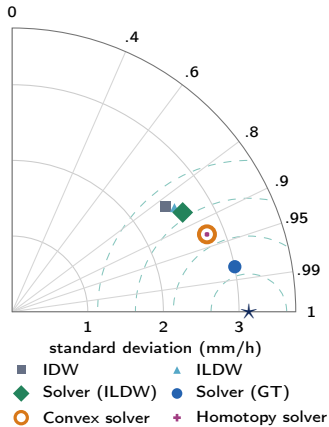
Solver(GT) is an optimistic diagnostic, initialized from the reference field.

Paired comparison: Convex Solver vs IDW

Metric	Better direction	Convex wins	IDW wins	Patch-level evidence
RMSE	lower	100	0	All patches favor Convex Solver
Absolute bias	lower	78	22	Reject null: $p = 4.667 \times 10^{-7}$
Pearson correlation	higher	100	0	All patches favor Convex Solver

Paired, one-sided t-test on 100 patches for absolute-bias difference $d_P = |\mathrm{Bias}_{P,\mathrm{Convex}}| - |\mathrm{Bias}_{P,\mathrm{IDW}}|$, with $H_0 : \mathbb{E}[d_P] \geq 0$.

Taylor diagram: aggregate across 100 patches



$\theta = \arccos \rho$: correlation;

F: SD (mm/h). Star: reference $(0, G)$. Dashed arcs: centered RMSE (mm/h).

- Each point summarizes 100 equally weighted patches.
- For patch k, g_k is the ground-truth rain map and f_k its reconstruction.
- Center each map by its own mean; compute $G_k = \mathrm{SD}(g_k)$, $F_k = \mathrm{SD}(f_k)$, and $\rho_k = \mathrm{corr}(g_k, f_k)$.

$$G = \sqrt{\tfrac{1}{100} \textstyle\sum_k G_k^2},$$

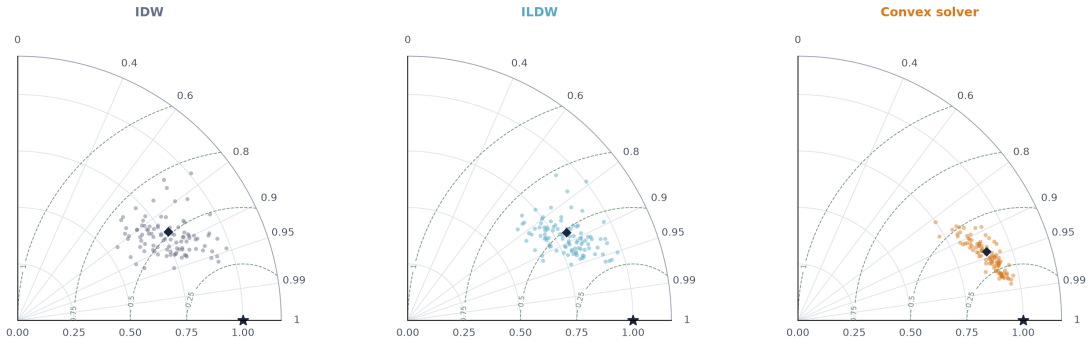
$$F = \sqrt{\tfrac{1}{100} \textstyle\sum_k F_k^2},$$

$$C = \tfrac{1}{100} \textstyle\sum_k G_k F_k \rho_k,$$

$$\rho = \frac{C}{GF},$$

$$(\theta, r) = (\arccos \rho,\ F).$$

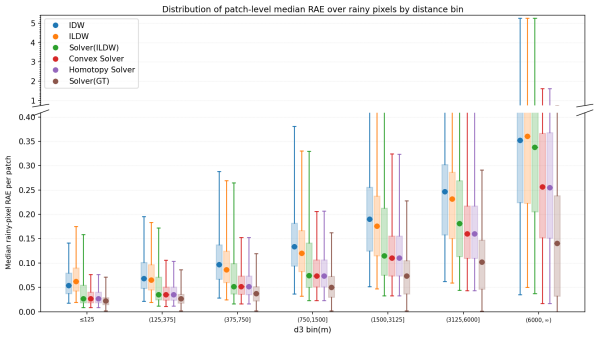
Taylor diagrams by reconstruction method



100 dots per method (one per patch) · black star: reference · black diamond: pooled moments · dashed arcs: centered RMSE

Each dot is one patch; the black diamond summarizes the 100 patches.

Reconstruction error over rainy pixels

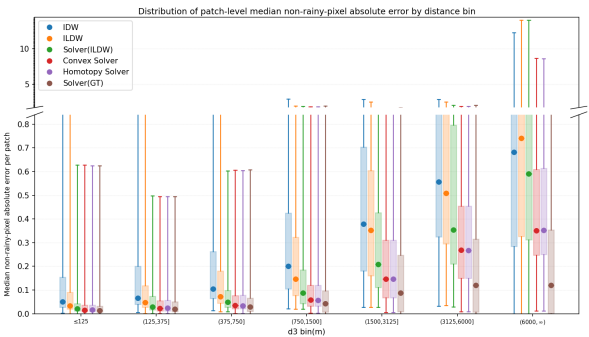


Mean wet pixels per patch
Total: 238,922

d_3 bin (m)	Pixels	% total
≤ 125	4,267	1.8
125–375	16,738	7.0
375–750	26,810	11.2
750–1500	51,989	21.8
1500–3125	88,839	37.2
3125–6000	45,929	19.2
> 6000	4,350	1.8

Wet pixels ($R \geq 0.6$ mm/h); relative absolute error by distance to the third-closest link.

Reconstruction error over dry pixels



Mean dry pixels per patch
Total: 30,569

d_3 bin (m)	Pixels	% total
≤ 125	564	1.8
125–375	2,261	7.4
375–750	3,389	11.1
750–1500	6,447	21.1
1500–3125	10,776	35.3
3125–6000	6,197	20.3
> 6000	936	3.1

Dry pixels ($R < 0.6$ mm/h); absolute error by distance to the third-closest link.

Wet/dry classification at 0.6 mm/h

Method	False-positive rate (of reference-dry pixels)	False-negative rate (of reference-wet pixels)
IDW	59.8%	0.3%
ILDW	56.2%	0.4%
Solver (ILDW init.)	46.3%	0.4%
Convex solver	40.0%	0.5%
Homotopy solver	40.1%	0.5%
Solver (GT init.)	33.1%	0.3%

On average, only **10.5%** of reference pixels are dry ($R < 0.6$ mm/h); the benchmark is strongly rain-dominated.

FPR = predicted wet among reference-dry pixels; FNR = predicted dry among reference-wet pixels.

End-to-end reconstruction runtime

Solver	Mean $\pm$ SD (s)	Median (s)	Range (s)	Optimizer time* (s)
IDW	7.53 ± 2.43	7.10	4.01–14.18	—
ILDW	18.69 ± 6.20	18.25	8.67–38.19	—
Convex solver	76.03 ± 27.08	71.31	32.76–153.12	57.00 ± 22.35

All times are per patch (100 patches); optimizer time is reported for the Convex Solver.

Optimizer time covers the solver call; end-to-end time also includes loading, setup, diagnostics, and saving. Sequential runs on an Apple M2 MacBook Air (8-core CPU, 8 GB RAM).

Conclusions and outlook

On our **synthetic benchmark**, Convex Solver improves map-level scores over IDW.

Does that advantage persist with real network measurements? We have not tested that yet.

1. Validate in practice

Use measured CML signal levels and compare with IDW.

2. Test other seasons

Compare maps across seasons: the winter benchmark is largely stratiform, while summer may include more convective rain.

3. Learn rainfall structure

Our current model uses explicit regularization, not a data-learned rainfall prior. Use rainfall maps to learn spatial patterns, while still requiring estimates to match the measured link attenuation.

A plausible CML-only use case: data-sparse African cities

Ouagadougou, Burkina Faso: the clearest documented example.

Evidence	Reported detail
CML rainfall map	100 unique paths; 5-min, 0.55-km grid (2020)
Gauge use	One reference gauge for evaluation, not a map input
Signal-data access	$>$ 1,000 links nationwide at 1-min sampling (2020–21); about 300 around Ouagadougou
Other cities	Niamey: 190 links; Douala: 227 links
Observation gap	Radar data not operationally available in Sahel; gauge networks sparse
Counts not reported	Comparable citywide gauge totals and radar footprints

Use case: reconstruct local rain fields from link attenuation when radar is unavailable and gauges are too sparse to resolve storm patterns.

References | Early attenuation research

Ryde, J. W. (1946). "The attenuation of centimetre radio waves and the echo intensities resulting from atmospheric phenomena." *J. Inst. Electr. Eng.*, 93(1), 101–103. doi:10.1049/JI-3A-1.1946.0029

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Data sources

EURADCLIM precipitation

Overeem, A. et al. (2023). "EURADCLIM: the European climatological high-resolution gauge-adjusted radar precipitation dataset." *Earth Syst. Sci. Data*, 15, 1441–1464. doi:10.5194/essd-15-1441-2023.

Dataset used: Overeem et al. (2025), EURADCLIM 1-hour accumulations, Version 3.0, KNMI Radar Team. DOI 10.21944/1rxx-ev62.

Commercial microwave-link data

Overeem, A., Walraven, B., Leijnse, H., & Uijlenhoet, R. (2024). *Four-year Commercial Microwave Link Dataset for the Netherlands*. 4TU.ResearchData, Version 1. doi:10.4121/BE252844-B672-471E-8D69-27269A862EC1.

The benchmark uses the dataset's link geometry and frequencies only. The original received signal levels (RSLs) are not used.

Figure and software credits

Rain-gauge photo. NOAA / National Weather Service (credit carried over from the earlier deck; source URL unavailable).

Weather-radar figures. FAA (2022), *Aviation Weather Handbook*, FAA-H-8083-28, Chapter 15.

Patch-location map. Footprints plotted from benchmark metadata; basemap: OpenStreetMap contributors / CARTO.

4TU network map. Link geometry: Overeem et al. (2024), 4TU.ResearchData; basemap: OpenStreetMap contributors / CARTO.

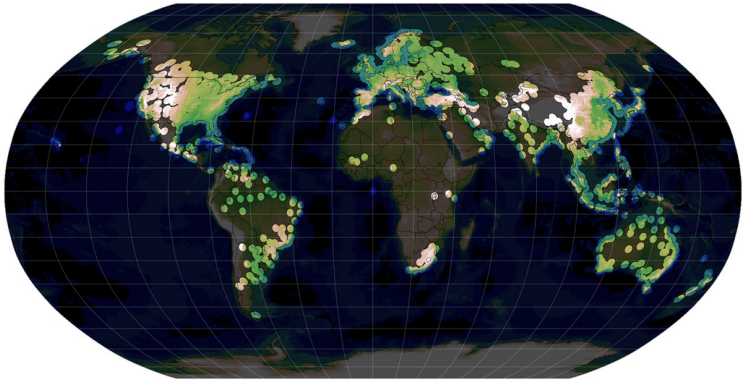
Rainfall/link example. EURADCLIM rainfall field with 4TU link geometry.

Generated illustrations. Cellular-network/rain and term graphics created or edited with OpenAI ChatGPT from author references.

Nonconvexity schematic. Based on collaborator-provided material.

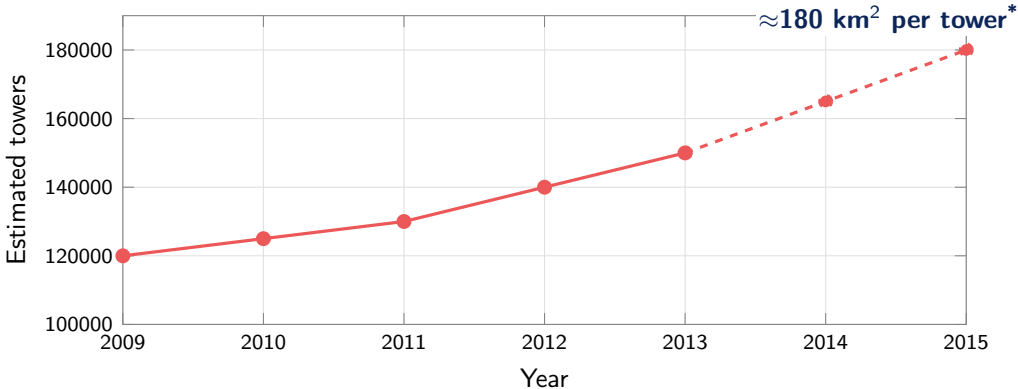
Beamer theme. SimplePlus by Pin-Yen Huang (2025), Unlicense. project page.

Global weather-radar coverage



Source: Saltikoff et al. (2019), Fig. 1, *Bulletin of the American Meteorological Society*, 100(9), 1739–1752. doi:10.1175/BAMS-D-18-0166.1
Method: Estimated radar illumination assumes a 200 km range for every radar; locations are based mainly on the WMO Radar Database (accessed March 2019). The database does not include every operational radar.

Africa's mobile-tower network was growing

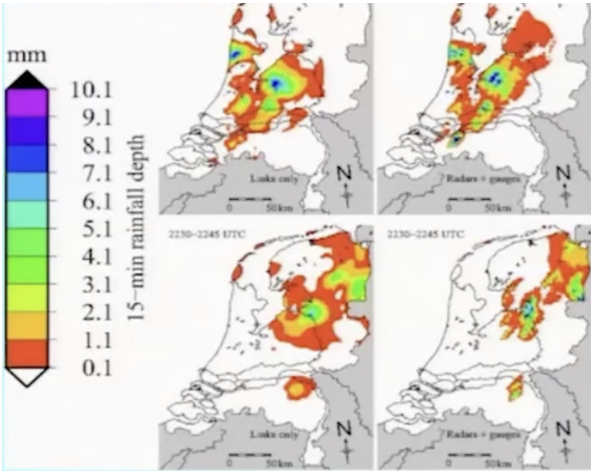


TowerXchange estimated 150,000 towers in 2013; its 2015 forecast was 180,000 (dashed).

*Area/count ratio, not coverage: 29.65 million km² (Africa) ÷ 165,000 towers (2014 forecast) ≈ 180 km²/tower. Area: UN Data.

Source: TowerXchange, Issue 10 (October 2014), Fig. 4, "African tower industry achieves launch velocity." TowerXchange article.

Reconstructing rainfall from cellular links



15-min rainfall, 10 Sep 2011: 20:30–20:45 UTC (top), 22:30–22:45 (bottom); links (left), radar + gauges (right).

Where? The Netherlands.

How? Convert link attenuation to path-averaged rainfall, then use ordinary kriging from link midpoints to map the field.

Comparison? Link-only estimates are checked against an independent radar-and-gauge product.

Source
Overeem, Leijnse, and Uijlenhoet (2013), *PNAS* 110(8), Fig. 3. doi:10.1073/pnas.1217961110